Beyond Onset Detection: Integrating ASR-Based Content Validation into Spoken-Word Production Annotation

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INTRODUCTION & MOTIVATION

The Verification Gap in Current Practice

Voice onset time (VOT) —latency between stimulus onset and start of the spoken response — central dv. in spoken-word production research.

Chronset (Roux et al., 2017) is the standard for automated VOT detection (r = .97 vs. trained raters) Chronset detects *when* speech starts — not *what* was said

Self-corrections, wrong words noise artifacts pass through undetected, manual verification remains necessary

Automation Meets Annotation

No existing pipeline integrates acoustic onset detection with ASR-based content validation for semi-automated review

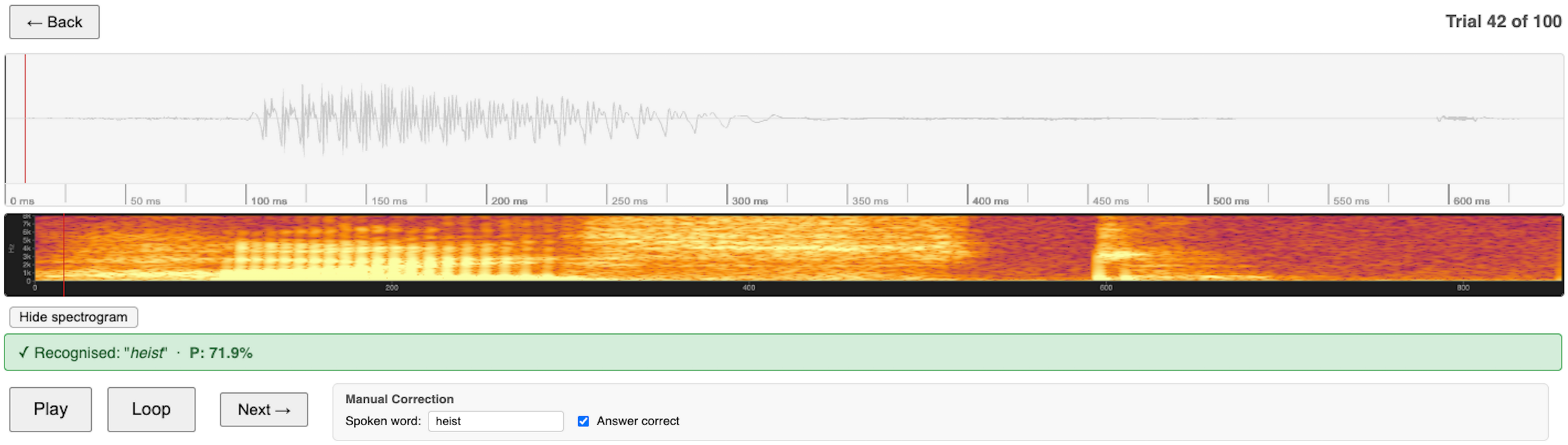
Our approach: route uncertain trials to a human reviewer — speeding up annotation while preserving data quality

① Detect onset automatically via Chronset
② Verify via ASR + phonological matching & correct onset if necessary
③ Human review for all or flagged / unresolved cases

The Pipeline

Step 1 — Onset Detection: Chronset voice onset timestamps from WAV files. Accurate for *when* speech starts — no information about *what* is said.
Step 2 — ASR Verification: faster-whisper (large-v3-turbo, INT8) transcribes each WAV segment. Matching pipeline resolves the transcription against the target word:
① **Exact match** — Whisper transcript equals target word (UK→US spelling correction)
② **Homophone resolution** — CMUdict ARPAbet cross-referenced with a curated homophone list. Example: *weigh* → [W EY1] = *way* → [W EY1]
③ **Phonetic similarity** — Levenshtein distance on CMUdict phonemes. Catches near-homophones not in the curated list (e.g. accent-shifted vowels)
Step 3 — Annotation Interface: Each trial displays:

- **Whisper transcription** with match banner, with probability (token log-probability), helps to judge whether mismatch was valid pronunciation variant
- **Waveform + spectrogram with draggable VOT marker** – can be manually adjusted, loop playback to support perceptual verification
- **Response override** — text field for target word correction; trial navigation via keyboard shortcut minimises response latency during annotation



RESULTS — Validation Across Two English Datasets: Auditory English Lexicon Project (AELP; *N* = 600, 6 speaker varieties) | Massive Auditory Lexical Decision (MALD; *N* = 1,000, 1 speaker)

93.3%

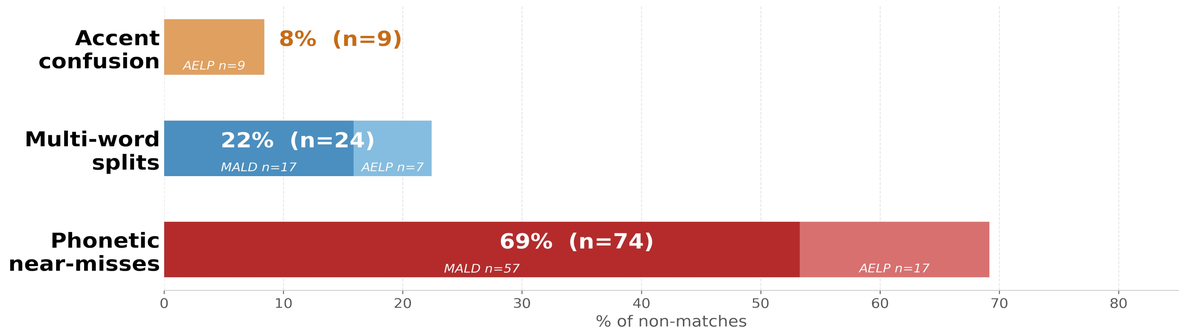
MALD: 926 / 1000, 1 speaker (Canadian English)
AELP: 567 / 600, 6 speaker varieties

1493	107	7
Auto-verified	Flagged (~ 6,7%)	Speakers

Annotated Workflow

- MALD1: 92.6% (926/1,000) auto-verified · AELP: 94.5% (567/600) auto-verified. Expert review targets flagged cases only — phonetic near-misses and multi-word splits
- Annotators review first the 107 flagged cases → focusing expert time on genuine ambiguities (accent confusion, multi-word splits, noise)

NON-MATCH BREAKDOWN — Combined AELP + MALD (107 cases)



Accent confusion (8%) — Whisper maps non-native vowels to nearest token (e.g. halt → "hold", detour → "die Tür"); AELP only
Multi-word splits (22%) — Whisper splits polysyllabic words (e.g. annum→"and im", un+popularity→"and popularity")
Phonetic near-misses (69%) — Whisper transcribes close but not exact (e.g. ban→ben, bomber→bummer)

ACHIEVEMENT & FUTURE DIRECTIONS

Pipeline Achievement

Integrated pipeline combining automated VOT detection, ASR verification, and guided annotation interface
3-stage matching: CMUdict ARPAbet reverse index → curated homophone list (482 groups, 1,016 words)
offline, dialect-aware (en-us/en-gb), zero-dependency annotation UI

✓ Low-latency CPU inference (faster-whisper INT8)
✓ Correction UI for VOT

Future Directions

➔ **Multiple language extension**: same pipeline applicable to different languages
➔ **Prefix/suffix recovery**: Whisper misses weak unstressed onset syllables — "asocial" → "social", "defile" → "file (known Whisper limitation; currently mitigated via 300 ms silence prepend)
➔ **direct integration of Chronset** via Matlab
➔ Lab vs. online recording robustness (SNR comparison)

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